RR-S-550D
February 7, 1980
SUPERSEDING
Int. Fed. Spec. RR-S-00550C(NAVY-SH)
March 3, 1978 and
Fed. Spec. RR-S-550B
February 11, 1974

FEDERAL SPECIFICATION

SOCKETS, WIRE ROPE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers sockets for wire ropes used in rigging, hoisting, towing, stowage, excavating, logging and stump removing equipments, and marine applications.

1.1.1 Federal specification coverage. Federal specifications do not include all types, finishes and classes of the commodities indicated by the title of the specification, or which are commercially available, but are intended to cover the types, finishes, and classes which are suitable for Federal Government requirements.

1.2 Classification.

1.2.1 Types, finishes, and classes. Wire rope sockets shall be of the following types, finishes, and classes as specified (see 6.2.1):

Types:

A - Open sockets.
B - Closed sockets.
C - Wedge sockets, open.
D - Bridge sockets, open.
E - Bridge sockets, closed.

Finishes:

1 - Bare (uncoated).
2 - Zinc-coated.

Classes:

1 - Not magnetic particle inspected.
2 - Magnetic particle inspected.

1.3 Sizes. Wire rope sockets shall be specified by wire rope diameters (see 3.3 and 6.2.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

FSC 4030

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Federal Specifications:

FF-N-836 - Nut: Square, Hexagon, Cap, Slotted, Castle, Knurled, Welding and Single Ball Seat.
FF-P-386 - Pins, Cotter (Split).
PPP-B-585 - Box, Wood, Wirebound.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Box, Wood, Nailed and Lock Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-665 - Box, Paperboard, Metal Edges and Components.

Federal Standards:

FED-STD-H28 - Screw Thread Standards for Federal Services.
FED-STD-66 - Steel, Chemical Composition and Hardenability.
FED-STD-123 - Marking for Shipment, (Civil Agencies).
FED-STD-151 - Metal, Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston; New York; Philadelphia; Washington, DC; Atlanta; Chicago; Kansas City, MO; Fort Worth; Houston; Denver; San Francisco; Los Angeles; and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-P-116 - Preservation-Packaging, Methods of.
MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads for 40" x 48" Pallets.
MIL-STD-163 - Steel Mill Products, Preparation for Shipment and Storage.
MIL-STD-271 - Nondestructive Testing Requirements for Metals.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; With Appropriate Test Methods.

Military Publication:

Naval Sea Systems Command (NAVSEA)
0900-LP-003-8000 - Metals, Surface Inspection Acceptance Standards.

(Copies of Military Specifications and Standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM):
A 153 - Zinc Coating (Hot-Dip) on Iron and Steel Hardware.
A 370 - Mechanical Testing of Steel Products, Methods and Definitions for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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Uniform Classification Committee, Agent:
Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee Agent, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

National Motor Freight Traffic Assiciation, Inc., Agent:
National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

3. REQUIREMENTS

3.1 Material.

3.1.1 The socket body (basket, wedge, and bridge) shall be made of carbon steel of compositions 1030, 1035 or 1038, as specified in FED-STD-66. The steel shall have a minimum tensile strength (normalized) of 70,000 pounds per square inch (lb/in^2) and a minimum elongation of 15 percent.

3.1.1.1 Eyebolts, U-bolts, nuts and pins shall be of the same steel compositions or of any other steel having identical tensile strength.

3.2 Fabrication.

3.2.1 Wire rope sockets, types A and B. These sockets shall be steel, drop forged, without any welding.

3.2.2 Wire rope sockets, type C. These sockets shall be steel, cast, and the wedges shall be drop forged, cast or cut from plate.

3.2.3 Wire rope sockets, types D and E. Bridge bowls in these sockets shall be steel, cast.

3.2.3.1 Eyebolts. Eye bolts in open bridge sockets, type D, shall be steel, forged, without any welding.

3.2.3.2 U-bolts. U-bolts in closed bridge sockets, type E, shall be made from steel bar stock, rolled or forged, at the contractor's option.

3.2.3.3 Nuts. The eyebolts and U-bolts shall carry at each end either single, finished, thick, or double heavy semi-finished hexagon nuts at the contractor's option, and shall be as specified in FF-N-836.

3.2.3.4 Threads. Threads shall be of the Unified or American National Coarse-Thread Series, class 2A for bolts and class 2B for nuts, in accordance with FED-STD-H28. The threads in the eyebolts and U-bolts shall be rolled or cut at the contractor's option.

3.2.4 Pins. Pins shall be made from bar stock, rolled or forged.

3.2.4.1 Cotter pins. Cotter pins shall conform to type B pins of FF-P-386.

3.2.5 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and shall be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.3 Constructional and dimensional requirements. Wire rope sockets are sized by the diameter of wire rope on which the sockets will be used.

3.3.1 Wire rope open sockets, type A. These sockets shall be of the shape shown on figure 1, and of dimensions given in table I, as specified.

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TABLE I. Wire rope open sockets, type A (all dimensions in inches). 1/ 2/

Size (rope diameter)	Basket				Jaws							Pin		Cotter pin diameter	Approximate weight, lbs.
	Length	Diameters			Sides		Eyes 3/					Diameter	Length		
		Top	Bottom		Width	Thickness	Outside diameter	Thickness	Width between eyes	C/L eye to					
			o.d.	i.d.						Basket top	Jaw top				
	J	H	G	F	K	E	M	N	C	L	B	D	O	P	
1/4	2-1/4	1- 9/16	11/16	3/8	3/4	1/4	1-5/16	5/16	11/16	1-9/16	3/4	11/16	1-3/4	3/16	1.1
5/16 - 3/8	2-1/4	1-11/16	13/16	1/2	13/16	5/16	1-1/2	7/16	13/16	1-3/4	7/8	13/16	2-1/16	3/16	1.3
7/16 - 1/2	2-1/2	1- 7/8	15/16	9/16	1	3/8	1-7/8	1/2	1	2	1- 1/16	1	2-7/16	3/16	2.3
9/16 - 5/8	3	2- 1/4	1- 1/8	11/16	1- 1/4	7/16	2-1/4	9/16	1- 1/4	2-1/2	1- 1/4	1- 3/16	2-7/8	1/4	3.8
3/4	3-1/2	2- 5/8	1- 1/4	13/16	1- 1/2	1/2	2-5/8	5/8	1- 1/2	3	1- 7/16	1- 3/8	3-1/4	1/4	6
7/8	4	3- 1/8	1- 1/2	15/16	1- 3/4	5/8	3-1/8	3/4	1- 3/4	3-1/2	1- 3/4	1- 5/8	3-7/8	5/16	10
1	4-1/2	3- 5/8	1- 3/4	1- 1/8	2	11/16	3-3/4	7/8	2	4	2- 1/16	2	4-1/2	3/8	15.5
1-1/8	5	4	2	1- 1/4	2- 3/8	3/4	4-1/8	1	2- 1/4	4-1/2	2- 5/16	2- 1/4	5	3/8	22
1-1/4 - 1-3/8	5-1/2	4- 5/8	2- 1/4	1- 1/2	2- 3/4	15/16	4-3/4	1- 1/8	2- 1/2	5	2-11/16	2- 1/2	5-5/8	7/16	32
1-1/2	6	5- 1/4	2- 3/4	1- 5/8	3	1	5-3/8	1- 3/16	3	6	3- 1/8	2- 3/4	6-3/8	1/2	46
1-5/8	6-1/2	5- 1/2	3	1- 3/4	3- 1/4	1- 1/8	5-3/4	1- 5/16	3	6-1/2	3- 1/4	3	6-5/8	1/2	55
1-3/4 - 1-7/8	7-1/2	6- 3/8	3- 1/8	2	3- 7/8	1- 5/16	6-1/2	1- 9/16	3- 1/2	7	3- 3/4	3- 1/2	7-5/8	1/2	85
2 - 2-1/8	8-1/2	7- 3/8	3- 3/4	2- 1/4	4- 1/4	1- 7/16	7	1-13/16	4	9	4	3- 3/4	8-3/4	1/2	125
2-1/4 - 2-3/8	9	8- 1/4	4	2- 1/2	4- 3/8	1- 5/8	7-3/4	2- 1/8	4- 1/2	10	4- 1/2	4- 1/4	10	1/2	165
2-1/2 - 2-5/8	9-3/4	9- 1/4	4- 1/2	2- 7/8	4- 5/8	1- 7/8	8-1/2	2- 3/8	5	10-3/4	5	4- 3/4	11	1/2	252

1/ All dimensions (except D and G) under 4 inches to have a tolerance plus and minus 1/8-inch. All dimensions (except D and G) 4 inches and over to have a tolerance plus and minus 1/4-inch. Dimension D shall have a tolerance plus 0 minus 1/32-inch in uncoated sockets and plus and minus 1/32-inch in zinc-coated sockets. Dimension E, socket sizes 1-3/4 inches and larger, shall have a tolerance of plus and minus 1/4-inch. Dimension G is minimum.

2/ All dimensions are to be measured at the high point of all drafts, except at the basket where both lineal and diameter dimensions are to the outside edge of the basket.

3/ The pin hole (eye) diameter in sockets supplied with pins shall be in accordance with table II.

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3.3.1.1 Pins. Type A sockets shall be supplied with pins. The diameters of the pin hole (eye) shall be as given in table II.

TABLE II. Diameters of pin hole (eye).

Socket size	Pin hole (eye) diameter: maximum over the diameter of the pin
Inches	Inch
1/4 - 5/8	1/16
3/4 - 1-3/8	3/32
1-1/2 - 2-1/8	1/8
2-1/4 - 2-3/8	1/4

3.3.1.2 Grooves. The basket in these sockets shall have one or more circumferential grooves on the inside surface, as shown on figure 1. The number and depth of the grooves shall be as specified in table III.

TABLE III. Number and the size of the grooves in wire rope sockets.

Socket size	Grooves	
	Number	Depth (approxi-mately)
Inches		Inch
1/4 to 3/4, inclusive	1	1/16
7/8 to 1-1/2, inclusive	2	1/8
1-5/8 and over	3	3/16

3.3.2 Wire rope closed sockets, type B. These sockets shall be of the shape shown on figure 2 and of dimensions given in table IV, as specified.

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TABLE IV. Wire rope closed sockets, type B (all dimensions in inches). 1/ 2/

Size (rope diameter)	Basket				Loop (bail)						Approximate weight, lbs.
	Length	Diameters			Width	Depth	Opening thickness				
		Top	Bottom				Width	Length	Sides	Top curve	
			O.d.	I.d.							
	J	H	G	F	C	K	D	L	E	B	
1/4	2-1/4	1-11/16	11/16	3/8	1- 1/2	1/2	13/16	1-3/4	5/16	1/2	0.7
5/16 - 3/8	2-1/4	1-11/16	13/16	1/2	1-11/16	11/16	15/16	2	3/8	5/8	1.1
7/16 - 1/2	2-1/2	1- 7/8	15/16	9/16	2	7/8	1- 1/8	2-1/4	7/16	11/16	1.5
9/16 - 5/8	3	2- 3/8	1- 1/8	11/16	2- 5/8	1	1- 3/8	2-1/2	5/8	13/16	3
3/4	3-1/2	2- 3/4	1- 1/4	13/16	3	1- 1/4	1- 5/8	3	11/16	1- 1/16	4.5
7/8	4	3- 1/4	1- 1/2	15/16	3- 5/8	1- 1/2	1- 7/8	3-1/2	7/8	1- 1/4	7
1	4-1/2	3- 3/4	1- 3/4	1- 1/8	4- 1/8	1- 3/4	2- 1/4	4	15/16	1- 3/8	11
1-1/8	5	4- 1/8	2	1- 1/4	4- 1/2	2	2- 1/2	4-1/2	1	1- 1/2	16
1-1/4 - 1-3/8	5-1/2	4- 3/4	2- 1/4	1- 1/2	5	2- 1/4	2- 3/4	5	1- 1/8	1- 5/8	22
1-1/2	6	5- 1/4	2- 3/4	1- 5/8	5- 3/8	2- 1/2	3- 1/8	6	1- 1/8	1-15/16	28
1-5/8	6-1/2	5- 1/2	3	1- 3/4	5- 3/4	2- 3/4	3- 1/4	6-1/2	1- 1/4	2- 1/8	36
1-3/4 - 1-7/8	7-1/2	6- 3/8	3- 1/8	2	6- 3/4	3	3-17/32	7-9/16	1- 1/2	2- 3/16	58
2 - 2-1/8	8-1/2	7- 3/8	3- 3/4	2- 1/4	7- 5/8	3- 1/4	3-25/32	8-9/16	1- 5/8	2- 7/16	80
2-1/4 - 2-3/8	9	8- 1/4	4	2- 1/2	8- 1/2	3- 5/8	4- 9/32	9-1/2	1- 3/4	2- 5/8	105
2-1/2 - 2-5/8	9-3/4	9- 1/4	4- 1/2	2- 7/8	9- 1/2	4	5- 1/2	10-5/8	2	3- 1/8	140

1/ All dimensions, except B, D, G, and L, under 4 inches to have a tolerance plus and minus 1/8-inch. All dimensions, except B, D, G, and L, 4 inches and over, to have a tolerance plus and minus 1/4-inch. Dimensions B, D, G, and L are minimum. Dimension E, socket sizes 1-3/4 inches and larger, to have a tolerance of plus and minus 1/4-inch.

2/ All dimensions are to be measured at the high point of all drafts except at the basket, where both lineal and diameter dimensions are to the outside edge of the basket.

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3.3.2.1 Grooves. The basket in these sockets shall have one or more circumferential grooves on the inside surface as shown on figure 2 and specified in table III.

3.3.3 Wire rope open wedge sockets, type C. These sockets shall be of the shape shown on figure 3 and of dimensions specified in table V. The baskets may be provided at the contractor's option, with ways for directing the sliding action of the wedge made with corresponding gibs. The wedge shall be grooved to suit the wire rope.

TABLE V. Wire rope wedge sockets, type C.

Size (rope diameter)	Minimum dimensions (inches)				Approximate weight
	Pin hole center to socket end	Distance between jaws	Thickness of jaws	Pin hole diameter, minimum	
	A	B	C	D	
Inches					Pounds
3/8	4-3/4	5/8	3/8	13/16	2.5
1/2	5-1/2	5/8	1/2	1- 1/16	3.5
5/8	6-7/8	1-1/4	1/2	1- 3/16	5
3/4	7-1/2	1-3/8	5/8	1- 1/4	9
7/8	9	1-1/2	3/4	1- 5/8	15
1	9-3/4	1-5/8	7/8	1- 5/8	20
1-1/8	10-5/8	1-5/8	1	1- 5/8	23
1-1/4	11-3/4	1-3/4	1-1/8	2- 1/8	32
1-3/8	11-3/4	1-3/4	1-1/8	2- 1/8	32
1-1/2	13-1/4	2-1/2	1-1/4	3- 1/8	52
1-5/8	13-1/4	2-1/2	1-1/4	3- 1/8	52

3.3.3.1 Type C sockets shall be supplied with pins. The diameters of the pins shall be smaller than the diameters of corresponding eye holes, specified in table V, by 1/32-inch in uncoated sockets and by 1/16-inch in coated (galvanized) sockets.

3.3.4 Wire rope open bridge sockets, type D. These sockets shall be of the construction shown on figure 4, and of dimensions specified in table VI.

TABLE VI. Wire rope open bridge sockets, type D.

Size (rope diameter)	Minimum dimensions (inches)							Approximate weight in pounds	
	Depth: zinc bowl	Eye bolts							
		Diameter	Length	Take up	C/C Distance	Eye hole diameter	Opening between eyes	Assembled socket only	Additional foot of bolt length
	A	B	C	D	E	F	G		
Inches									
5/8	3-1/4	7/8	20	8	3-1/2	1-3/16	1-1/4	25	5.5
3/4	3-3/4	1	21	9	4	1-3/8	1-1/2	30	5.5
7/8	4-1/4	1-1/8	23	9	4-1/2	1-1/2	1-3/4	37	7
1	5	1-1/4	25	11	5	1-3/4	2	55	7
1-1/8	5-1/2	1-3/8	27	12	5-1/2	2	2-1/4	80	10
1-1/4	6	1-1/2	27	12	6	2-1/4	2-1/4	110	12
1-3/8	6-1/2	1-3/4	28	12	6-1/2	2-1/2	2-3/4	150	16
1-1/2	7	2	31	15	7-1/8	2-1/2	3	190	21
1-5/8	7	2	32	15	7-1/8	2-3/4	3-1/4	240	21
1-3/4	8	2-1/4	36	18	8-1/4	3	3-1/2	290	27
1-7/8	8	2-1/2	38	18	8-1/4	3	3-3/4	340	33
2	9	2-1/2	38	18	9-1/2	3-1/4	3-3/4	400	33

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TABLE VI. Wire rope open bridge sockets, type D. - Continued

Size (rope diameter)	Minimum dimensions (inches)							Approximate weight in pounds	
	Depth: zinc bowl	Eyebolts						Assembled socket only	Additional foot of bolt length
		Diameter	Length	Take up	C/C Distance	Eye hole diameter	Opening between eyes		
	A	B	C	D	E	F	G		
Inches									
2-1/8	10	2-3/4	40	18	10-1/4	3-3/4	4	460	40
2-1/4	10	2-3/4	40	18	10-1/4	3-3/4	4	520	40
2-3/8	11	3	42	18	11-1/4	4-1/4	4-1/2	600	48
2-1/2	11	3	42	18	11-1/4	4-1/2	4-1/2	700	48
2-5/8	12	3-1/4	42	18	12	4-3/4	5	800	56
2-3/4	12	3-1/4	42	18	12	4-3/4	5	900	56
2-7/8	12-1/2	3-1/2	42	18	13-1/2	5	5-1/2	1000	66
3	13	3-1/2	44	18	13-1/2	5-1/4	5-1/2	1100	66

3.3.4.1 Pins. Type D sockets shall be supplied with pins. The diameters of the pins shall be smaller than the diameters of corresponding eye holes, specified in table VI, by 1/32-inch in uncoated sockets and by 1/16-inch in coated (galvanized) sockets.

3.3.5 Wire rope closed bridge sockets, type E. These sockets shall be of the construction shown on figure 5 and of dimensions as specified in table VII.

TABLE VII. Wire rope closed bridge sockets, type E.

Size (rope diameter)	Minimum dimensions (inches)					Approximate weight, in lbs.	
	Depth of zinc bowl	U-bolts				Assembled socket only	Additional foot of bolt length
		Diameter	Length	Takeup	C/C distance		
	A	B	C	D	E		
Inches							
5/8	3-1/4	7/8	15	8	3-1/2	16	5.5
3/4	3-3/4	1	16	9	4	20	5.5
7/8	4-1/4	1-1/8	17	9	4-1/2	28	7
1	5	1-1/4	18	9	5	38	7
1-1/8	5-1/2	1-3/8	22	12	5-1/2	55	10
1-1/4	6	1-1/2	24	12	6	75	12
1-3/8	6-1/2	1-3/4	26	12	6-1/2	97	16
1-1/2	7	2	30	15	7-1/8	125	21
1-5/8	7	2	30	15	7-1/8	145	21
1-3/4	8	2-1/4	33	15	8-1/4	185	27
1-7/8	8	2-1/2	33	18	8-1/4	220	33
2	9	2-1/2	38	18	9-1/2	290	33
2-1/8	10	2-3/4	39	18	10-1/4	320	40
2-1/4	10	2-3/4	40	18	10-1/4	380	40
2-3/8	11	3	42	18	11-1/4	450	48
2-1/2	11	3	42	18	11-1/4	525	48
2-5/8	12	3-1/4	42	18	12	580	56
2-3/4	12	3-1/4	42	18	12	630	56
2-7/8	12-1/2	3-1/2	44	18	13-1/2	690	66
3	13	3-1/2	44	18	13-1/2	750	66

3.4 Galvanizing. Wire rope sockets with finish 2 shall be galvanized (zinc-coated) by the hot-dip galvanizing process. The sockets on nonmilitary orders shall be galvanized in accordance with good commercial practice, whereas those on military orders shall be galvanized in accordance with the requirements of ASTM A 153.

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3.4.1 Eyebolts and U-bolts and nuts in sockets with finish 2 shall be galvanized after threading or tapping. Nuts may be retapped after galvanizing.

3.5 Marking for identification. Each socket on nonmilitary orders shall be plainly marked or branded with its size, manufacturer's name or trademark and on military orders in addition with the letters "U.S.".

3.6 Magnetic particle inspection. All type A and B, finish 1, class 2 sockets shall be completely magnetic particle inspected in accordance with MIL-STD-271. These sockets shall comply with class 3 of NAVSEA 0900-LP-003-8000. Indications in excess of class 3 shall be cause for rejection of that socket.

3.7 Workmanship. All sockets shall be properly shaped without sharp edges at openings, free from flaws, seams, fins, slivers, cracks and other injurious defects which may affect their serviceability and handling. The zinc coating, (finish 2) on wire rope sockets shall be adherent, smooth, free from uncoated spots, injurious lumps, blisters, dross or flux.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contractor or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lot. An inspection lot shall consist of all class 1 and class 2 sockets of the same type and finish produced under the same conditions and offered for inspection at one time.

4.2.2 Sampling. Statistical sampling and inspection shall be in accordance with MIL-STD-105. The acceptable quality level (AQL) is 2.

4.2.3 Sampling for chemical analysis. One sample shall be taken from several sockets selected at random from the lot. The sample shall consist of not less than 2 ounces of fine, clean (free from oil, dirt, grit or other foreign material) drillings, millings, or shavings, properly packed, marked, and sent to the laboratory designated by the contracting activity.

4.2.3.1 If the sockets are made of steel from a heat which has been previously sampled and found in conformance with this specification, the sampling of 4.2.3 may be waived.

4.2.4 Sampling for tensile test. Four representative samples shall be selected at random from each lot of sockets or from the steel to be used in the manufacture of each lot of sockets.

4.3 Inspection (for dimensional, workmanship and marking (branding) requirements). Each of the sample sockets, selected in accordance with 4.2.2 shall be measured for compliance with the specified dimensional requirements and visually inspected for workmanship and marking or branding. If the number of defective sockets in the sample is less or equal to the acceptance number specified in 4.2.2 for the size of the lot under inspection, the lot represented by the sample shall be accepted insofar as the dimensional, marking (branding) or workmanship requirements are concerned. If the number of defective sockets is equal or greater than the rejection number specified in 4.2.2 for the size of the lot under inspection, the lot represented by the sample shall be rejected.

4.4 Tests.

4.4.1 Chemical analysis test. A chemical analysis shall be made on the sample prepared in accordance with 4.2.3. The analysis shall be made by a wet chemical or spectrographic method. In case of dispute, the analysis by the wet chemical process shall be the basis for acceptance. Failure of the sample to conform to the chemical composition, specified by 3.1.1, shall be cause for rejection of the lot.

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4.4.2 Tensile test. The preparation of the test specimens and the testing methods shall be conducted in accordance with the requirements of ASTM A 370. Failure of one test specimen to conform to the required tensile strength (see 3.1.1), shall be cause for rejection of the lot.

4.5 Reinspection and retest.

4.5.1 Reinspection. A rejected lot may be resubmitted for reinspection after the contractor or manufacturer, being informed of the reasons for rejection, has inspected each socket in the lot for the deficiency(ies) noted and has removed or reworked all nonconforming units. One reinspection only shall be permitted.

4.5.2 Retests. Retests, when requested by the manufacturer or contractor, shall be permitted in accordance with the regulations of FED-STD-151. Not more than one retest shall be permitted.

4.6 Magnetic particle inspection. All type A and B, finish 1, class 2 sockets shall be magnetic particle inspected in accordance with 3.6.

4.7 Inspection of packaging. Sample packages and packs and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PREPARATION FOR DELIVERY

(The packaging requirements specified herein apply only for direct Government acquisitions.)

5.1 General. Wire rope sockets of one type, finish and size shall be packaged, packed, or palletized in the quantity specified in table VIII.

TABLE VIII. Quantities of sockets per case (container).

Socket size	Quantity of sockets per case	Socket size	Quantity of sockets per case	Socket size	Quantity of sockets per case
Inch		Inches		Inches	
1/4	100 1/	3/4	20	1-5/8	4
5/16	100 1/	7/8	10	1-3/4	3
3/8	100 1/	1	10	1-7/8	3
7/16	50	1-1/8	8	2	2
1/2	50	1-1/4	5	2-1/8	2
9/16	40	1-3/8	5	2-1/4	2
5/8	30	1-1/2	4	2-3/8	2

1/ Packaged 10 sockets per carton and 10 cartons per case.

5.2 Preservation-packaging. Preservation-packaging shall be level A or C, as specified (see 6.2.1).

5.2.1 Level A. Wire rope sockets shall be cleaned process C-1 and unit protected method I of MIL-P-116. Contact preservative shall conform to type P-19 or MIL-P-116. Cartons (see footnote 1, table VIII) shall conform to class domestic boxes of PPP-B-665 or PPP-B-636, with box selection at the option of the contractor. For 7/16 size and larger, sockets shall be packed as specified in 5.3.

5.2.2 Level C. Preservation-packaging of wire rope sockets shall afford protection against corrosion, deterioration or physical damage during shipment from the supply source to the first receiving activity for immediate use. The contractor's normal preservation-packaging methods may be utilized when such meet the requirements of this level.

5.3 Packing. Packing shall be level A, B, or C, as specified (see 6.2.1).

5.3.1 General requirements for levels A and B. Shipping containers shall be of a minimum weight and cube consistent with the requirements of this specification. Anchoring, blocking, bracing and cushioning methods and materials shall be in accordance with MIL-STD-1186, except that shredded newspaper, excelsior, and similar materials which have not been chemically refined for noncorrosiveness are unacceptable unless they are enclosed in a sealed waterproof, greaseproof barrier material. For level A shipments, wood and wood-cleated boxes shall have case liners conforming to MIL-L-10547.

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5.3.2 Levels A and B. Sockets preserved-packaged as specified (see 6.2.1) shall be packed for the level specified (see 6.2.1) in any one of the containers as specified in table IX.

TABLE IX. Container selection.

Specification	Container	Application (style, type or class)		Use criteria gross weight maximum (pounds) 1/			
		Level A	Level B	Under 100	100-250	250-1000	Over 1000
PPP-B-636	Box, shipping, fiberboard	Weather resistant 3/	Domestic	2/	2/	No	No
PPP-B-585	Box, wood, wirebound	Class 3	Class 1	4/	Yes	2/	No
PPP-B-601	Box, wood, cleated-plywood	Overseas type	Domestic type	4/	Yes	Yes	No
PPP-B-621	Box, wood, nailed and lock-corner	Class 2 overseas	Class 1 domestic	4/	Yes	2/	No
MIL-STD-163	5/	5/	5/	No	4/	4/	Yes 6/
MIL-STD-147	Pallets 40" x 48"	6/	6/	No	No	No	Yes

1/ Containers with gross weight exceeding 200 pounds shall have wood skids applied in accordance with the applicable container specification or appendix thereto, or if no such requirement exists, skids as specified in PPP-B-621 shall be applied.

2/ Maximum gross weight, container plus contents, shall not exceed the applicable specification requirements for the style, type or class container selected for the level of packing to be applied.

3/ For Army only, fiberboard containers shall be prohibited as exterior shipping containers under level A. When used, fiberboard containers shall be over-packed in other optional wood containers specified.

4/ For weights under 100 pounds, container may be used; however, lighter weight, minimum cube and less expensive fiberboard containers should be given consideration.

5/ Table VI and figure 2, bar boxes of MIL-STD-163 may also be used for the weights specified.

6/ When specified (see 6.2.1), sockets may be palletized in accordance with MIL-STD-147 and softwood lumber is acceptable as substitute for hardwood.

5.3.2.1 Shipping containers shall be closed and strapped in accordance with the applicable specification or appendix thereto, with method V for weather-resistant and method I for domestic closure applying for PPP-B-636 containers.

5.3.3 Level C. Sockets, preserved-packaged as specified (see 6.2.1), shall be packed in containers acceptable to the common carrier which will insure safe delivery at destination in a satisfactory condition at the lowest applicable rate. Containers, packing or method of shipment shall comply with Uniform Freight or National Motor Freight Classification Rules or Regulations or other carrier rules as applicable to the mode of transportation.

5.4 Marking.

5.4.1 Military agencies. In addition to any special marking required by the contract (see 6.2.1), interior packages, exterior shipping containers or pallets shall be marked in accordance with MIL-STD-129.

5.4.2 Civil agencies. In addition to any special marking required by the contract (see 6.2.1), interior packages and exterior shipping containers shall be marked in accordance with FED-STD-123.

6. NOTES

6.1 Intended use.

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6.1.1 Wire rope sockets, types A, B, D, and E. These sockets are for permanent attachment to wire ropes. The end of a wire rope is threaded through the lower opening in the basket or the bridge, the wires are untwisted and broomed-up, the core removed and after thorough pickling, washing and thorough drying of wires, molten zinc is poured into the basket or bridge bowl. After the solidification of zinc, the wire rope and the socket are firmly and permanently assembled until zinc is remelted. This method of assembly of wire ropes with sockets is used wherever the full strength of the wire rope has to be utilized when needed. The bridge sockets are used in constructions where tightening of ropes from time to time is involved.

6.1.2 Wire rope sockets, type C. These sockets are used for quick, temporary or field assembly requirements and where untwisting of the wire rope is undesirable. The ultimate assembly strength of this type of socket is about 70 percent of the wire rope strength.

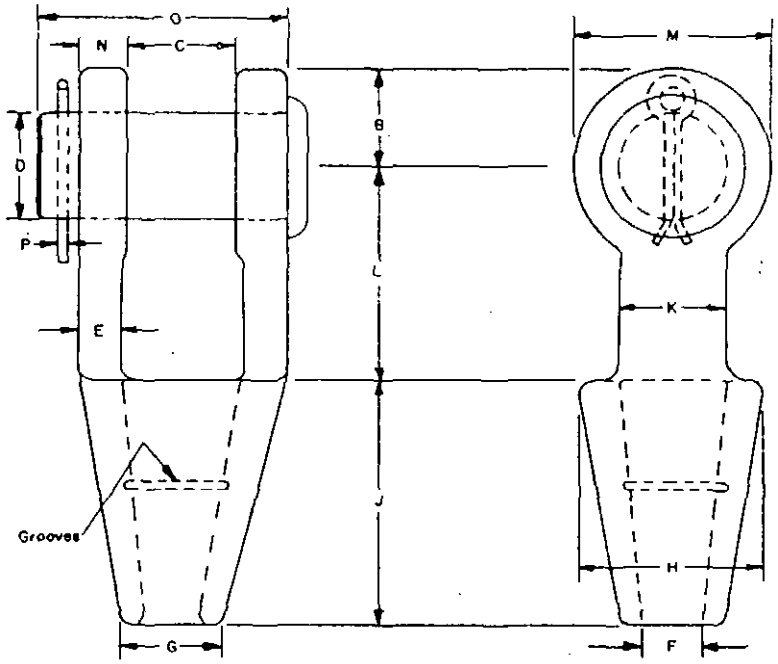
6.1.3 Wire rope sockets, classes 1 and 2. Class 2 sockets are to be used where magnetic particle inspection is required such as in lifting nuclear equipment. Class 1 sockets are to be used for all other applications.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, finish, and class of sockets required (see 1.2).
(c) Size of sockets required (see 1.3).
(d) Number of sockets required.
(e) Levels of preservation-packaging and packing required (see 5.2, 5.3, 5.3.2, and 5.3.3).
(f) When sockets shall be palletized (see 5.3.2, footnote 6).
(g) Special marking required (see 5.4.1 and 5.4.2).

6.3 American National Standard M11.1, "Wire Ropes for Mines", may be consulted as a safety standard for the proper usage of items covered by this specification.



SH 10646

FIGURE 1. Wire rope open socket, type A.

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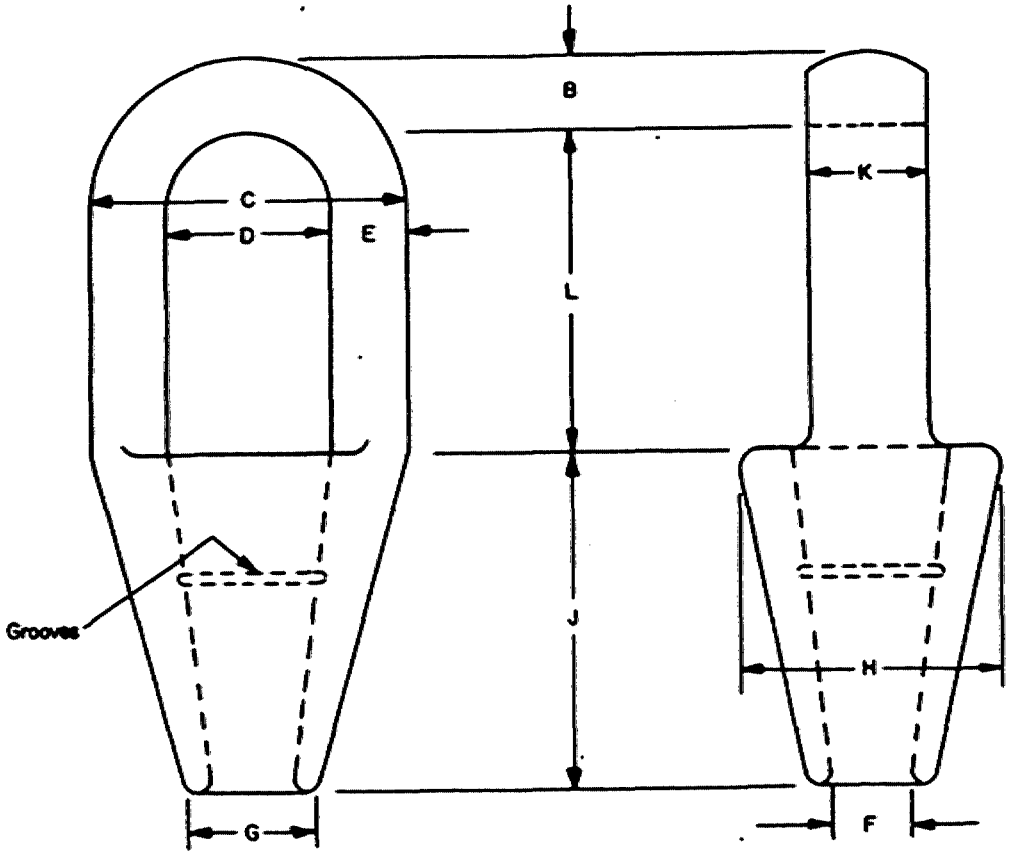
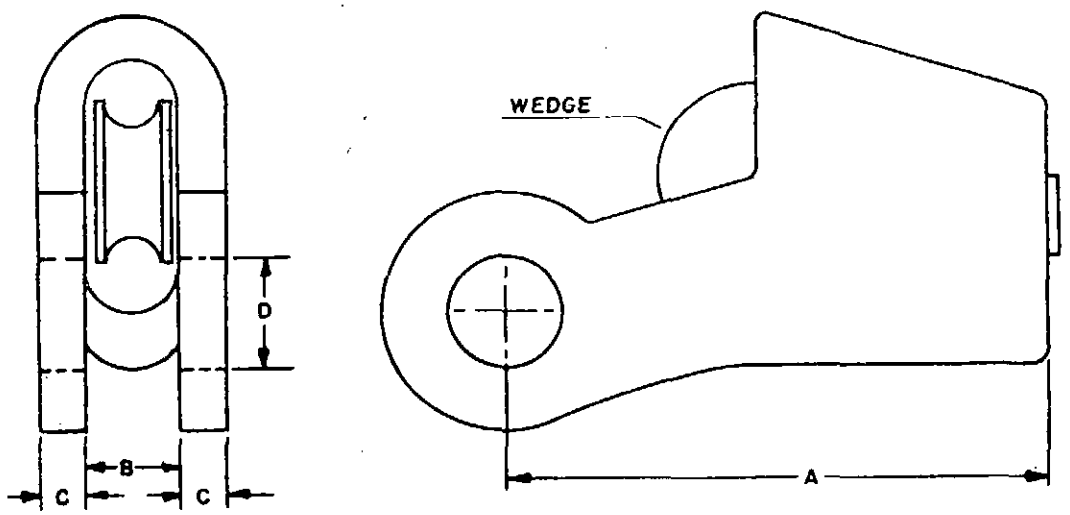


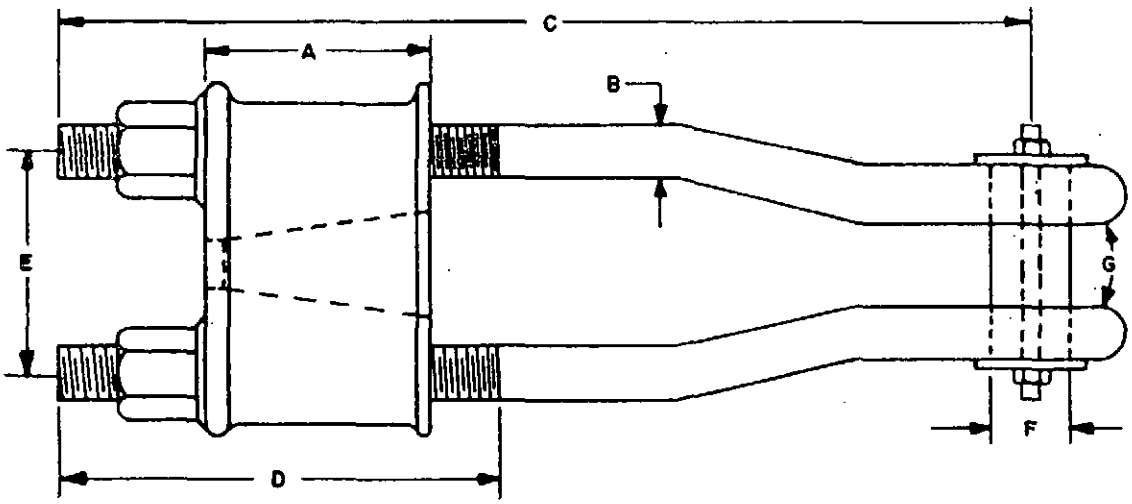
FIGURE 2. Wire rope closed socket, type B.

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SK 10648

FIGURE 3. Wire rope wedge socket, type C.



SK 10649

FIGURE 4. Wire rope open bridge socket, type D.

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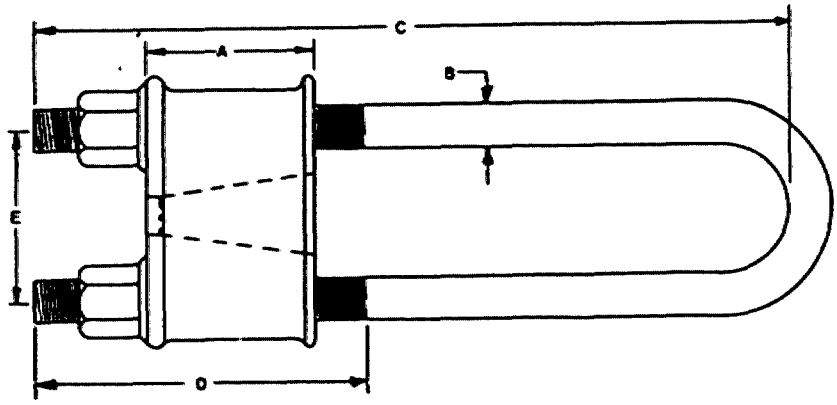


FIGURE 5. Wire rope closed bridge socket, type E.

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Navy - SH

Review activities:
Army - EL
Navy - YD
DSA - IS

User activities:
Navy - MC, CG

Civil Agency Coordinating Activity:
GSA - FSS

Preparing Activity:
Navy - SH

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